CUSTOMER SUPPORT HANDBOOK

FiberNet Broadband Network Issues Troubleshooting

A practical Level 1 / Level 2 support guide for FTTH broadband, Wi-Fi, router, ONT and customer-premises network incidents

Document purpose	Standardize diagnosis, restoration, customer communication and escalation for broadband incidents.
Primary audience	Customer support agents, NOC/service desk staff, field technicians and team leads.
Technology scope	FTTH/GPON or XGS-PON access, ONT/ONU, Ethernet router, dual-band Wi-Fi, DNS and common application
Guiding principle	Check service status first, then optical link, WAN session, LAN/Wi-Fi and finally device/application causes.

Important: This is a vendor-neutral operational guide. Exact LED names, optical thresholds, PPPoE/VLAN values and reset procedures must be aligned with the ISP's approved configuration.

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1. Support Scope and Troubleshooting Model

Use a layered approach so that agents do not repeatedly reboot equipment or change settings before identifying the failure domain. A broadband incident normally belongs to one of five layers: provider outage, optical access, WAN/IP session, home LAN/Wi-Fi, or end-device/application.

Recommended diagnostic sequence

- **Step 1 - Confirm impact:** one device, all devices, Wi-Fi only, wired only, one website/app, or complete outage.
- **Step 2 - Check known outage/maintenance:** validate service area, OLT/PON alarms and active maintenance notices.
- **Step 3 - Inspect ONT/ONU:** power, PON/optical registration, LOS/alarm state and Ethernet link.
- **Step 4 - Inspect router WAN:** WAN link, IP assignment, PPPoE/DHCP status, VLAN profile and DNS.
- **Step 5 - Test LAN:** Ethernet first where possible; then evaluate Wi-Fi signal, band, channel and interference.
- **Step 6 - Isolate device/application:** test another device, browser, DNS resolver or service.
- **Step 7 - Restore and verify:** confirm browsing, speed, latency and customer use case before closing.

Fault-domain quick map

Symptom	Most likely domain	First check
LOS red / blinking	Fiber path / optical access	Patch cord, connector, splitter/PON status
PON normal, no internet on all devices	WAN/session/router	WAN IP, PPPoE/DHCP, outage
Ethernet works, Wi-Fi fails	Wireless LAN	SSID, band, signal, channel, router radio
Only one device fails	End device	IP/DNS, adapter, VPN/proxy, forget/rejoin
Only one app/site fails	Application/DNS/path	Alternate site, DNS, service status
Slow at peak hours on all devices	Capacity/congestion or Wi-Fi	Wired speed test + utilization

Do not factory-reset the ONT/router as an early step. A reset can remove ISP provisioning, VLAN, voice and Wi-Fi settings and may convert a recoverable issue into a configuration incident.

2. FiberNet / FTTH Architecture Basics

Understanding the service path helps the agent identify where a failure can occur. In a typical FTTH service, traffic flows from the ISP core to an OLT, through passive optical splitters and distribution fiber, to the customer's ONT/ONU. The ONT then hands Ethernet to a router or integrated residential gateway.

Logical service path

Layer	Component	What can fail	Typical evidence
Provider/core	Internet edge, BNG/BRAS, DNS	Outage, routing, authentication	Many customers affected
Access	OLT/PON port	Port alarm, capacity, provisioning	PON registration issue
Optical distribution	Feeder/drop fiber, splitter	Cut, bend, dirty connector	LOS / low optical power
Customer optical	ONT/ONU	Power, registration, hardware	LED alarm / no Ethernet
WAN	Router WAN/PPPoE/DHCP	No IP, auth, VLAN	WAN disconnected
LAN	Ethernet/Wi-Fi	Cable, DHCP, RF interference	Local-only symptoms
Device/app	PC/mobile/TV/app	DNS, VPN, OS, app	Single-device/service impact

Key terms agents should know

- **OLT:** Optical Line Terminal at the provider side that serves multiple PON subscribers.
- **ONT/ONU:** Customer-premises optical device converting fiber to Ethernet/voice/Wi-Fi.
- **PON:** Passive Optical Network. GPON and XGS-PON are common access technologies.
- **LOS:** Loss of Signal. Usually indicates the ONT is not receiving usable optical light.
- **Optical power:** Receive/transmit light level measured in dBm. Acceptable ranges are ISP/vendor specific.
- **PPPoE:** Username/password based WAN session used by some ISPs; others use DHCP/IPoE.
- **VLAN:** Logical network tag used to separate internet, voice or IPTV services.

When speaking to customers, avoid unnecessary acronyms. Translate the diagnosis into simple language, for example: 'Your router has power, but the fiber terminal is not receiving a usable signal from the network.'

3. Intake, Customer Verification and Triage

Good troubleshooting begins with precise intake. Record what is failing, when it started, whether it is intermittent, and whether the customer has changed any equipment, cabling or settings.

Minimum intake checklist

- Verify account/service identifier using the organization's approved authentication process.
- Confirm installation address/service circuit and registered contact information.
- Capture start time, frequency and whether the issue is continuous or intermittent.
- Ask whether all devices are affected and whether Ethernet and Wi-Fi behave differently.
- Ask which LEDs are lit on the ONT and router; request exact color and whether steady or blinking.
- Check for recent power outage, electrical work, router relocation, cable movement or renovation.
- Check active outage/maintenance tools before asking the customer to perform disruptive steps.

Severity classification

Priority	Example	Handling
P1 / Critical	Widespread service outage, major aggregation/core impact	Immediate NOC/incident escalation
P2 / High	Business/critical subscriber down, repeated PON impact	Priority technical escalation
P3 / Normal	Single residential line down or severe degradation	Standard troubleshooting/SLA
P4 / Low	Wi-Fi optimization, configuration query, non-service-affecting request	Advisory / scheduled support

Useful opening questions

1) 'Is the internet unavailable on every device or only one?' 2) 'Can you check whether the fiber box shows a red LOS/alarm light?' 3) 'Does a device connected by Ethernet also have the problem?' 4) 'Did this start after a power cut or any cable/router change?'

Record observations before changing anything. A clear before/after state is valuable when the case is escalated.

4. ONT/ONU LED Diagnosis

LED labels differ by manufacturer, but the pattern usually reveals whether the problem is power, optical registration, WAN handoff or LAN connectivity. Use the ISP's device-specific guide whenever available.

LED / state	Typical meaning	Support action
Power off	No power to ONT	Check adapter, socket, UPS and power switch
Power on, PON off	Not registered / no usable optical link	Check LOS and optical path
PON solid, LOS off	Optical registration normally healthy	Proceed to WAN/router checks
LOS red/blinking	Optical signal missing or below threshold	Inspect fiber; check PON outage; escalate if needed
LAN off	No Ethernet link to router/device	Reseat/replace cable; verify correct port
LAN blinking	Ethernet traffic present	Does not prove internet is working
Internet off/red	WAN/session failure on integrated gateway	Check IP/PPPoE/VLAN/service status
TEL/Phone abnormal	Voice registration issue	Check voice provisioning after internet/PON state

Safe physical checks

- Confirm the ONT power adapter is firmly connected and is the correct rated adapter.
- Ensure the thin fiber patch cord is not sharply bent, crushed under furniture or pulled tight.
- Do not look into a fiber connector or optical port. Invisible laser light may be present.
- Do not repeatedly unplug optical connectors. Dirty connector end faces can create attenuation.
- If the ISP permits customer reseating, handle the connector by its body and avoid touching the ferrule/end face.
- Confirm Ethernet is connected from the designated ONT LAN port to the router WAN/Internet port.

Escalate optical faults when LOS persists after basic physical checks, the customer reports a damaged drop cable, multiple nearby subscribers are affected, or remote optical readings/OLT alarms are abnormal.

5. No Internet - All Devices

This is the primary end-to-end outage workflow. The objective is to distinguish provider/optical failure from router/WAN failure without making unnecessary configuration changes.

Decision workflow

- **1.** Check outage/maintenance systems. If confirmed, inform the customer and attach the case to the incident.
- **2.** Check ONT power and optical LEDs. LOS/alarm indicates an optical path issue; do not focus on Wi-Fi.
- **3.** If PON is healthy, verify the Ethernet link between ONT and router WAN.
- **4.** Check whether the router has a valid WAN IP and gateway. A blank/0.0.0.0 address indicates session/IP assignment failure.
- **5.** For PPPoE, verify session status and approved credentials/provisioning; do not disclose stored passwords.
- **6.** Reboot in the correct order only if appropriate: power off router, restart ONT if policy permits, wait for PON registration, then power on router.
- **7.** Test a known website and DNS resolution. If IP connectivity works but names fail, follow the DNS workflow.
- **8.** If authorized, test directly from ONT Ethernet only when the ISP design supports it and security policy allows.

Common outcomes

Finding	Likely cause	Next step
LOS active	Fiber/optical fault	Field/NOC escalation
PON healthy, WAN no IP	DHCP/PPPoE/VLAN/provisioning	Remote session/config check
WAN IP present, no browsing	DNS/routing/firewall	Ping/path/DNS isolation
Internet returns after router reboot	Router state/session issue	Monitor recurrence; firmware/hardware if repeated
Direct test works, router fails	Router/configuration	Repair/reprovision/replace router

After restoration, confirm at least two destinations and one customer-priority use case such as work VPN, streaming or messaging.

6. Intermittent Connection / Frequent Disconnects

Intermittent faults require evidence over time. Avoid closing the case simply because the connection is working during the call. Capture timestamps and correlate them with optical, WAN and power events.

Isolation checklist

- Determine whether disconnects affect all devices simultaneously. If only Wi-Fi devices drop, focus on RF/Wi-Fi.
- Ask whether ONT LOS/PON LEDs change during the outage. Optical LED changes strongly indicate access instability.
- Check remote optical history, ONT deregistration events, PPPoE session drops and OLT alarms where tools allow.
- Check router uptime. Unexpected uptime resets can indicate power instability, overheating or hardware reboot.
- Inspect power adapters, extension boards and loose sockets. Prefer a stable wall outlet for testing.
- Check fiber bend radius and whether the cable is pinched by a door, cabinet or recent furniture movement.
- For WAN session drops with stable optical state, check PPPoE/DHCP lease/session logs and duplicate-session behavior.
- For Wi-Fi-only drops, compare 2.4 GHz and 5/6 GHz, signal strength and interference.

Evidence to capture

Evidence	Why it matters
Exact disconnect timestamps	Allows correlation with OLT, BNG, router and power logs
ONT LED state during failure	Separates optical from WAN/LAN causes
Router uptime / event log	Identifies reboot or WAN renegotiation
Wired vs Wi-Fi behavior	Separates ISP/WAN from wireless issue
Frequency and duration	Helps distinguish brief session flaps from sustained outages

If the issue recurs at predictable times, record the pattern. Scheduled electrical loads, RF interference, network congestion or maintenance may be time-correlated.

7. Slow Internet / Speed Complaints

Speed complaints must be measured under controlled conditions. A Wi-Fi test can be limited by radio conditions, device capability or router placement even when the broadband circuit is delivering the subscribed rate.

Standard speed-test method

- Confirm the subscribed plan and whether the complaint is download, upload, latency or all three.
- Pause cloud backup, streaming, game downloads and other heavy traffic on the home network.
- Use a capable computer connected by Gigabit/2.5G Ethernet where possible.
- Verify Ethernet link speed. A 100 Mbps negotiated link will cap throughput below 100 Mbps.
- Use the ISP-approved speed-test server or a nearby reputable server. Run multiple tests.
- Compare wired result with Wi-Fi result. If wired is healthy, treat the case as Wi-Fi optimization.
- For plans above 1 Gbps, confirm ONT, router, NIC, cable and test server support multi-gigabit throughput.
- Record time, server, ping, download and upload results.

Interpretation examples

Observed result	Likely explanation
Wired near plan rate; Wi-Fi low	Wireless signal/interference/device limitation
Wired and Wi-Fi both low	WAN capacity, link negotiation, traffic load or provider issue
Download good; upload poor	Upstream impairment/congestion/configuration
Speed good; apps feel slow	Latency, packet loss, DNS or application/server issue
Exactly ~90-95 Mbps	Likely 100 Mbps Ethernet link
Exactly ~900-950 Mbps on 1 Gbps	Normal practical Gigabit Ethernet ceiling

Do not promise exact headline speed over Wi-Fi. Actual Wi-Fi throughput varies with distance, walls, interference, channel width, client radio capability and concurrent usage.

8. High Latency, Packet Loss and Gaming/Voice Issues

A connection can have high bandwidth but poor responsiveness. Real-time applications such as gaming, voice and video meetings are sensitive to latency variation (jitter) and packet loss.

Diagnostic sequence

- Test from Ethernet first to remove Wi-Fi variability.
- Ping the local router/default gateway. Loss here indicates LAN/Wi-Fi/device problems.
- Ping or test the ISP first-hop/gateway if support tools permit. Loss beginning here may indicate access/provider issues.
- Test a stable public destination and compare latency with the local/ISP hop.
- Use traceroute/path diagnostics as supporting evidence, not as proof by itself; intermediate routers may de-prioritize ICMP.
- Check whether latency increases only while uploads/downloads are saturated. This can indicate queueing/bufferbloat.
- For gaming, distinguish game-server latency from ISP path latency; server region and peering matter.
- For voice/video, note jitter, packet loss and whether problems occur only on Wi-Fi.

Practical interpretation

Test	Healthy indication	Concern
Router ping	Very low and stable	Loss/spikes suggest local network issue
Wired public ping	Stable for destination/region	Repeated loss or large unexplained spikes
Wi-Fi vs wired	Similar under good signal	Wi-Fi much worse = RF/local issue
Under load	Moderate increase	Very large increase = queueing/congestion

When escalating, include the test method, whether Ethernet was used, destination, timestamps and packet-loss sample. A screenshot alone without context is weak evidence.

9. Wi-Fi Coverage and Weak Signal

Wi-Fi is a local radio network and is affected by distance, walls, floor slabs, metal, mirrors, appliances and neighboring networks. Fiber service quality and Wi-Fi coverage must be diagnosed separately.

Placement guidance

- Place the router centrally and in the open, not inside a closed cabinet.
- Keep it elevated and away from large metal objects, aquariums and dense concrete obstructions.
- Avoid placing the router immediately beside microwave ovens, cordless phone bases or other strong RF sources.
- For multi-floor or large homes, use a properly designed mesh/access-point system rather than a single high-power router.
- Use Ethernet backhaul for mesh/access points where practical for more consistent performance.

Band selection

Band	Strength	Trade-off	Best use
2.4 GHz	Longer reach / better wall penetration	Lower speed, more interference	Distant/IoT devices
5 GHz	Higher capacity and speed	Shorter range	Phones, laptops, streaming near AP
6 GHz	Clean spectrum / high capacity	Shortest reach; client support required	Modern devices near AP

Support checks

- Confirm the customer is connected to the intended SSID and access point.
- Compare signal and performance near the router versus the problem room.
- If separate SSIDs are available, compare 2.4 GHz and 5 GHz behavior.
- Check channel utilization/interference using approved diagnostics.
- For mesh, verify node status, backhaul quality and whether the client is attached to an appropriate node.

A Wi-Fi extender can increase coverage but may reduce throughput when it repeats traffic over the same radio. Mesh or wired access points are generally more predictable for larger homes.

10. Wi-Fi Connected but No Internet

The Wi-Fi icon only confirms that the device is associated with the local wireless network. It does not prove that the router has a working internet connection.

Troubleshooting workflow

- Check whether other devices on the same Wi-Fi can browse. If all fail, investigate WAN/ONT rather than the individual device.
- Open the router status page/support tool and confirm WAN IP, gateway and DNS information.
- If only one device is affected, toggle Wi-Fi off/on, then forget and rejoin the SSID if needed.
- Check whether the device received a valid private IP, gateway and DNS server. Self-assigned/APIPA addressing indicates DHCP failure.
- Disable VPN/proxy temporarily for diagnosis if customer policy permits.
- Test a known IP destination and then a hostname. IP works but hostname fails points to DNS.
- Check captive portal behavior on managed/public-style networks if applicable.
- Restart the affected device before making router-wide changes.

Common client IP clues

Observation	Meaning	Action
169.254.x.x	No DHCP lease received	Check DHCP/router association; reconnect
Valid private IP, no gateway	Incomplete DHCP/config	Renew lease; router DHCP check
Valid IP/gateway, DNS missing	DNS assignment issue	Check WAN/DHCP/DNS
Valid settings, one device only	Device/VPN/security stack	Reset adapter/network settings as last resort

Avoid asking the customer to change to public DNS as the first step. First confirm that the fault is actually DNS-related and follow the ISP's policy.

11. DNS and Website/Application Access Problems

DNS translates hostnames into IP addresses. A DNS issue can make the internet appear down even when IP connectivity is working. Conversely, one unavailable website does not automatically indicate a DNS fault.

Isolation process

- Confirm whether multiple unrelated websites fail or only one service.
- Test another device on the same connection.
- Check whether a known IP endpoint is reachable while hostname lookup fails.
- Use nslookup/dig or approved diagnostic tools to compare the configured resolver with an alternate resolver where policy permits.
- Flush the local DNS cache or restart the affected device when stale cache is suspected.
- Check browser secure-DNS/DoH settings, VPN clients, security software and parental-control filtering when only one device is affected.
- For a single site/app, check whether the service itself is unavailable or region/account restricted.
- Escalate suspected ISP DNS failure when multiple subscribers show consistent resolver failures.

Useful commands for trained support staff

Platform	Command	Purpose
Windows	ipconfig /all	View IP, gateway and DNS configuration
Windows	ipconfig /flushdns	Clear local DNS resolver cache
Windows	nslookup example.com	Test DNS resolution
Windows	ping <gateway>	Test local gateway reachability
Windows	tracert example.com	View IP path (supporting evidence)
macOS/Linux	nslookup or dig	Test DNS resolver response
macOS/Linux	traceroute	View network path

Commands should be used only by staff who understand the output. Never ask customers to run unknown scripts or download unapproved diagnostic utilities.

12. Ethernet and LAN Troubleshooting

Ethernet provides a useful baseline because it removes wireless interference. Many apparent broadband speed problems are caused by a damaged cable, incorrect port, or link negotiating at 100 Mbps instead of 1 Gbps or higher.

Physical checks

- Confirm the cable is connected from the ONT LAN port to the router WAN/Internet port.
- For client testing, connect the computer to a router LAN port, not the WAN port.
- Reseat both ends and inspect clips/connectors for damage.
- Swap with a known-good Cat5e or better cable for Gigabit testing.
- Check link LEDs and the negotiated speed shown by the router/NIC.
- Avoid unverified couplers, damaged wall jacks and very long temporary cables during baseline testing.

Negotiated link vs expected throughput

Link	Approximate practical ceiling	Interpretation
100 Mbps Ethernet	Below 100 Mbps	Will bottleneck plans above 100 Mbps
1 Gbps Ethernet	Roughly 900-950 Mbps	Normal for many 1 Gbps tests
2.5 Gbps Ethernet	Above 1 Gbps possible	Needed for many multi-gigabit plans
10 Gbps Ethernet	Multi-gigabit	Requires compatible end-to-end hardware

DHCP/LAN checks

- Verify the client receives an IP in the expected LAN subnet.
- Check for duplicate/static IP conflicts.
- Confirm the router DHCP pool is not exhausted.
- If a managed switch is present, check VLAN and port configuration before assuming ISP failure.
- If only one LAN port fails, test another port and document the hardware behavior.

A successful Ethernet baseline is one of the strongest ways to separate a broadband service issue from a Wi-Fi issue.

13. Router Reboot, Reset and Configuration Safety

Rebooting and resetting are different actions. A reboot restarts the device while preserving configuration. A factory reset erases configuration and should be used only when the ISP procedure specifically requires it.

Recommended reboot sequence

- Save or document any relevant status before rebooting.
- Power off the router.
- If policy allows and an ONT restart is needed, restart the ONT and wait until optical registration is stable.
- Power on the router and allow the WAN session to establish fully.
- Retest using the same method used before the reboot.

Factory reset - use only with authorization

Risk	Possible result
WAN profile erased	PPPoE/VLAN/IP settings may be lost
Wi-Fi configuration erased	SSID/password revert to defaults
Voice/IPTV profile erased	Additional services may stop
Remote management lost	Device may no longer be reachable by support systems
Custom customer settings lost	Port forwarding, reservations and parental controls removed

Configuration safety rules

- Do not change VLAN, WAN mode, MTU or authentication values without an approved service profile.
- Do not expose PPPoE passwords, admin passwords or Wi-Fi credentials in case notes.
- Back up configuration where supported before major changes.
- After a replacement or reset, validate internet, Wi-Fi and any bundled voice/IPTV service.
- Change only one major variable at a time so the result can be interpreted.

Factory reset is a last-resort recovery step, not a generic troubleshooting step.

14. Router, ONT and Power Hardware Issues

Hardware failures are less common than configuration or connectivity issues, but they should be suspected when symptoms persist after isolation or when there are clear power, thermal or port abnormalities.

Hardware warning signs

- Device repeatedly reboots even on a stable power source.
- Power adapter becomes unusually hot, makes noise, or has damaged cable/connector.
- ONT/router fails to power on with the approved adapter and known-good socket.
- One Ethernet port consistently fails while others work.
- Device becomes unstable after warming up and recovers after cooling.
- Wi-Fi radios disappear or cannot be enabled despite correct configuration.
- Physical damage, water ingress, burn marks or lightning/power-surge history.

Power troubleshooting

Check	Action
Wall socket	Test a known-good outlet; avoid overloaded extension boards
Adapter	Use only the manufacturer/ISP-approved voltage, polarity and current rating
UPS/inverter	Temporarily isolate if it is suspected, following electrical safety policy
Power cycle	Allow full boot/registration time before concluding failure
Surge/lightning	Escalate for inspection if multiple ports/devices were affected

Replacement criteria

Replace equipment only after the fault has been isolated according to support policy. Record model, serial/asset identifier, firmware, observed failure, tests performed and replacement result. For ONTs, ensure the new unit is provisioned/registered correctly before closure.

Never substitute a random power adapter simply because the plug fits. Incorrect voltage, polarity or insufficient current can damage equipment or create a safety risk.

15. Fiber Physical Layer and Optical Escalation

Optical troubleshooting must protect both service quality and safety. Customer support should perform only approved visual/remote checks; cleaning, splicing and optical measurement are normally field-technician tasks.

Common optical causes

- Drop fiber cut or crushed during construction, drilling or furniture movement.
- Excessive bend or kink causing attenuation.
- Dirty or damaged connector/end face.
- Loose connector or damaged coupler/adapter.
- Splitter/distribution fault affecting one or multiple subscribers.
- OLT/PON port issue or provisioning/registration problem.
- Optical level marginal and fluctuating due to physical degradation.

Remote/field data to collect

Data	Use
ONT registration state	Confirms whether the OLT sees the ONT
Rx/Tx optical level	Compares measured power with ISP/vendor thresholds
PON/OLT alarms	Identifies wider access fault
Neighbor impact	Helps isolate individual drop vs shared network
Last-seen / deregistration history	Supports intermittent optical diagnosis
Fiber route / recent civil work	Useful for probable physical damage

Optical power thresholds

Do not use a universal dBm pass/fail value in customer communication. GPON/XGS-PON optics and vendor classes have defined operating ranges, and the ISP may use tighter engineering thresholds. Compare readings with the approved network standard.

Safety: Never look into the end of an optical fiber or optical transceiver. Do not perform cleaning or splicing unless trained and equipped for fiber work.

16. Special Cases: VPN, Streaming, IPTV, Voice and Smart Devices

Some complaints affect only a particular service. First prove whether general internet connectivity is healthy, then isolate the application-specific path.

Use case	Checks
Work VPN	General internet, DNS, VPN client, credentials, corporate gateway status, MTU/path policy
Streaming	Wired/Wi-Fi throughput, device capability, app status, CDN/server region, concurrent usage
IPTV	Correct ISP port/VLAN, set-top box link, multicast/IGMP design if applicable
VoIP/landline	ONT voice LED, SIP/voice provisioning, handset/cable, dial tone, registration
Gaming	Wired latency/loss, NAT type, server region, UPnP/port policy, congestion under load
Smart TV/IoT	2.4 GHz support, WPA compatibility, DHCP, band steering, app/cloud availability
CCTV/cloud camera	Upload capacity, Wi-Fi signal, NAT/cloud reachability, device vendor service

VPN notes

- If normal browsing works but only the corporate VPN fails, do not immediately classify it as an ISP outage.
- Test another network only if permitted; it can help distinguish corporate VPN/client issues from the home path.
- Do not disable enterprise security controls or ask for corporate credentials.

Streaming notes

- Check whether buffering occurs on all streaming services or only one.
- Compare performance near the router or over Ethernet.
- 4K streaming can expose weak Wi-Fi even when basic browsing appears normal.

For bundled IPTV/voice, avoid changing VLAN or service profiles unless the exact provisioning design is known. Internet may work while another service is misconfigured.

17. Escalation to NOC / Field Technician

A strong escalation contains enough evidence for the next team to act without repeating the entire customer call. State the fault domain you suspect and the observations that support it.

Escalate to NOC when

- Multiple customers or a service area show similar symptoms.
- OLT/PON alarms, session platform issues, routing/DNS problems or capacity concerns are suspected.
- WAN authentication/IP assignment fails despite correct customer-premises state.
- Persistent packet loss/latency is reproduced on Ethernet and appears beyond the customer LAN.

Escalate to field support when

- LOS persists or optical readings are outside approved thresholds.
- Drop fiber, connector, wall outlet or ONT hardware is physically damaged.
- A field optical test, connector cleaning, splice or cable replacement is required.
- Remote steps cannot restore ONT registration or a hardware replacement is authorized.

Escalation template

Field	Example content
Customer impact	All devices offline since 18:20; no wired or Wi-Fi internet
ONT state	Power on; LOS blinking red; PON off
Router state	WAN down; Ethernet link to ONT present/absent as observed
Outage check	No known incident / incident ID if present
Actions taken	Power/cable inspection; approved reboot; no factory reset
Remote evidence	ONT unregistered; optical reading/alarm if available
Requested action	Check drop fiber/PON path and restore optical registration

Always include timestamps and the customer's current service state at the time of escalation.

18. Customer Communication, Case Notes and Closure

Technical accuracy and clear communication are equally important. Customers should understand what was found, what was changed, and what will happen next without being overwhelmed by internal terminology.

Communication principles

- Acknowledge the specific impact: 'I understand the connection is dropping on all devices every 10-15 minutes.'
- Explain each requested action and why it is needed.
- Do not blame the customer's equipment before isolation proves it.
- Give realistic restoration or escalation expectations based on the actual SLA/incident process.
- If an outage exists, avoid unnecessary home troubleshooting.
- Before closing, ask the customer to verify the service from their normal device/use case.

Case-note structure

Section	What to record
Problem	Symptoms, start time, affected devices/services
Environment	ONT/router model, wired/Wi-Fi, plan if relevant
Observations	LED state, WAN IP/session, optical state, link speed
Tests	Wired speed, ping/loss, DNS, alternate device
Actions	Cables reseated, reboot, configuration change, replacement
Result	Restored/not restored; verification results
Next step	Closed, monitor, NOC escalation, field visit

Closure checklist

- Service is restored or an escalation/incident is correctly assigned.
- Customer has tested the affected use case.
- Any temporary changes are reverted unless intentionally retained.
- Case notes contain no passwords or unnecessary personal information.
- Recurring faults are tagged/linked so trend analysis is possible.

Good notes reduce repeat troubleshooting and shorten future resolution time.

19. Quick Reference - Troubleshooting Matrix and Support Checklist

Use this page as a rapid call-side reference. Detailed procedures and ISP-specific values take precedence.

Customer symptom	Check first	Then
No internet, all devices	Outage + ONT LEDs	WAN IP/session, Ethernet
Red LOS	Fiber path / PON status	Optical escalation
PON normal, WAN down	ONT-router cable	DHCP/PPPoE/VLAN/provisioning
Wi-Fi connected, no internet	Other devices + WAN status	Client IP/DNS
Wi-Fi weak in rooms	Router placement/signal	Band/channel/mesh/AP
Slow speed	Wired controlled test	Link speed, utilization, provider capacity
Frequent drops	All vs Wi-Fi only	Optical/session/power logs
High ping/gaming lag	Ethernet latency/loss	Load, path, game server
One site/app fails	Other sites/devices	DNS/app/service status
Only one device fails	IP/DNS/VPN/proxy	Forget/rejoin or device network reset

Five-minute support checklist

- 1. Define impact: all devices, one device, Wi-Fi only, or one app.
- 2. Check outage/maintenance.
- 3. Read ONT Power/PON/LOS/LAN state.
- 4. Check router WAN IP/session and ONT-router Ethernet link.
- 5. Test Ethernet if speed, latency or Wi-Fi is disputed.
- 6. Test IP connectivity and DNS separately.
- 7. Reboot only when justified; avoid factory reset unless authorized.
- 8. Capture evidence and timestamps for intermittent faults.
- 9. Escalate optical/provider faults with complete notes.
- 10. Verify the customer's real use case before closure.

Recommended ticket tags

NO-INTERNET | LOS/OPTICAL | WAN-AUTH | DHCP/IP | DNS | SLOW-SPEED | LATENCY/LOSS | INTERMITTENT | WIFI-COVERAGE | ROUTER-HARDWARE | ONT-HARDWARE | IPTV | VOICE | VPN/APPLICATION

Final reminder: Troubleshoot from the provider edge toward the customer device, isolate one layer at a time, preserve configuration, and document evidence. This approach resolves issues faster and makes escalations actionable.

End of handbook